



STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

07 – Derivation of CG Shift Equations

This attachment derives the equations used in WA-OCEAN to solve for the center of gravity and to solve for forces to shift the center of gravity.

A center of gravity shift would be caused by changes in the distribution of weight within a module. At the time of analysis, the details causing a shift would be unknown. Thus, a CG shift is treated as a global effect.

Applied CG shifts are the sum of two components. The first is for a shift to match the weight report. The second is a shift when applying the CG envelope.

1.0 CALCULATION OF THE CENTER OF GRAVITY

The center of gravity is calculated using,

$$CG = \Sigma(\text{weight})(\text{distance}) / \Sigma(\text{weight}) \quad (\text{E.1})$$

The indicated summations combine the effect of each module component weight where,

CWi = weight of module component “i”

Lix, Liy, Liz = coordinate location of module component “i”

The module total weight (TW) is,

$$TW = \Sigma CWi \quad (\text{E.2})$$

The CG in each of the 3 axis directions (CGx, CGy, CGz) is computed from,

$$CGx = \Sigma (CWi)(Lix) / TW \quad (\text{E.3})$$

$$CGy = \Sigma (CWi)(Liy) / TW$$

$$CGz = \Sigma (CWi)(Liz) / TW$$

2.0 CALCULATING FORCES TO SHIFT THE CENTER OF GRAVITY

Once the CG is computed, module component weights can be replaced with the total weight applied at the center of gravity. Equation E.3 becomes,

$$CGx = (TW)(CGx) / TW \quad (\text{E.4})$$

$$CGy = (TW)(CGy) / TW$$

$$CGz = (TW)(CGz) / TW$$

The amount of desired CG shift is indicated as,

$\Delta CGx, \Delta CGy, \Delta CGz$ = dimension of shift from the unshifted center of gravity

A defined series of nodes is established such that they surround the CG. Each of the “Q” number of nodes “k” has a vertical load of “Wk” and a coordinate location of “Lkx”, “Lky”, and “Lkz”. The shifted CG is described using these definitions and equation E.4,



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$$CGx + \Delta CGx = [(TW)(CGx) + \Sigma (Wk)(Lkx)] / TW \quad (E.5)$$

$$CGy + \Delta CGy = [(TW)(CGy) + \Sigma (Wk)(Lky)] / TW$$

$$CGz + \Delta CGz = [(TW)(CGz) + \Sigma (Wk)(Lkz)] / TW$$

Expressing these equations with respect to the unshifted CG involves substituting for the coordinate locations,

$$CGx \rightarrow CGx - CGx \rightarrow 0 \quad Lkx \rightarrow Lkx - CGx \quad (E.6)$$

$$CGy \rightarrow CGy - CGy \rightarrow 0 \quad Lky \rightarrow Lky - CGy$$

$$CGz \rightarrow CGz - CGz \rightarrow 0 \quad Lkz \rightarrow Lkz - CGz$$

Substituting E.6 into E.5 results in,

$$0 + \Delta CGx = [(TW)(0) + \Sigma (Wk)(Lkx - CGx)] / TW \quad (E.7)$$

$$0 + \Delta CGy = [(TW)(0) + \Sigma (Wk)(Lky - CGy)] / TW$$

$$0 + \Delta CGz = [(TW)(0) + \Sigma (Wk)(Lkz - CGz)] / TW$$

$$\Delta CGx = \Sigma (Wk)(Lkx - CGx) / TW \quad (E.8)$$

$$\Delta CGy = \Sigma (Wk)(Lky - CGy) / TW$$

$$\Delta CGz = \Sigma (Wk)(Lkz - CGz) / TW$$

Substituting,

$$Dkx = Lkx - CGx \quad (E.9)$$

$$Dky = Lky - CGy$$

$$Dkz = Lkz - CGz$$

Results in,

$$\Delta CGx = \Sigma (Wk)(Dkx) / TW \quad (E.10)$$

$$\Delta CGy = \Sigma (Wk)(Dky) / TW$$

$$\Delta CGz = \Sigma (Wk)(Dkz) / TW$$



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The solution involves solving for each of the values of W_k . Because the CG shift is defined to be in the 3 axis directions, it follows that W_k should vary by positions (Dx, Dy, Dz) in the 3 axis directions. Thus let W_k be expressed generically as,

$$W_k = a_1 + a_2(Dkx) + a_3(Dky) + a_4(Dkz) \quad (E.11)$$

Where,

a_1 = a constant value

a_2 = coefficient for X axis dimension from CGx

a_3 = coefficient for Y axis dimension from CGy

a_4 = coefficient for Z axis dimension from CGz

Substituting E.11 into equation E.10 results in,

$$\Delta CGx = \Sigma [a_1 + (a_2)(Dkx) + (a_3)(Dky) + (a_4)(Dkz)] (Dkx) / TW \quad (E.12)$$

$$\Delta CGy = \Sigma [a_1 + (a_2)(Dkx) + (a_3)(Dky) + (a_4)(Dkz)] (Dky) / TW$$

$$\Delta CGz = \Sigma [a_1 + (a_2)(Dkx) + (a_3)(Dky) + (a_4)(Dkz)] (Dkz) / TW$$

Re-arranging equation E.12 results in,

$$(\Delta CGx)(TW) = \Sigma [(a_1)(Dkx) + (a_2)(Dkx)(Dkx) + (a_3)(Dkx)(Dky) + (a_4)(Dkx)(Dkz)] \quad (E.13)$$

$$(\Delta CGy)(TW) = \Sigma [(a_1)(Dky) + (a_2)(Dky)(Dkx) + (a_3)(Dky)(Dky) + (a_4)(Dky)(Dkz)]$$

$$(\Delta CGz)(TW) = \Sigma [(a_1)(Dkz) + (a_2)(Dkz)(Dkx) + (a_3)(Dkz)(Dky) + (a_4)(Dkz)(Dkz)]$$

Now the solution involves solving for the four alpha values. The fourth required equation is a summation of vertical (W_k) forces equaling zero, or $\Sigma (W_k) = 0$. This is expressed as,

$$0 = \Sigma [a_1 + a_2(Dkx) + a_3(Dky) + a_4(Dkz)] \quad (E.14)$$

Equations E.13 and E.14 can be converted to matrix form as, (E.15)

$$\begin{Bmatrix} (\Delta CGx)(TW) \\ (\Delta CGy)(TW) \\ (\Delta CGz)(TW) \\ 0 \end{Bmatrix} = \begin{bmatrix} \Sigma(Dkx) & \Sigma(Dkx)(Dkx) & \Sigma(Dkx)(Dky) & \Sigma(Dkx)(Dkz) \\ \Sigma(Dky) & \Sigma(Dky)(Dkx) & \Sigma(Dky)(Dky) & \Sigma(Dky)(Dkz) \\ \Sigma(Dkz) & \Sigma(Dkz)(Dkx) & \Sigma(Dkz)(Dky) & \Sigma(Dkz)(Dkz) \\ Q & \Sigma(Dkx) & \Sigma(Dky) & \Sigma(Dkz) \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{Bmatrix}$$



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This can be generically expressed as $\{CG\ TW\} = [D] \{\alpha\}$ (E.16)

And the solution is, $\{\alpha\} = [D]^{-1} \{CG\ TW\}$ or, (E.17)

$$\begin{Bmatrix} a1 \\ a2 \\ a3 \\ a4 \end{Bmatrix} = \begin{bmatrix} D11 & D12 & D13 & D14 \\ D21 & D22 & D23 & D24 \\ D31 & D32 & D33 & D34 \\ D41 & D42 & D43 & D44 \end{bmatrix} \begin{Bmatrix} (\Delta CGx)(TW) \\ (\Delta CGy)(TW) \\ (\Delta CGz)(TW) \\ 0 \end{Bmatrix}$$

The set of node forces is then solved using equation E.11.